

Mini-BEHAVIOR-Gran: Revealing U-Shaped Effects of Instruction Granularity on Language-Guided Embodied Agents

Anonymous ACL submission

Abstract

Instruction granularity is an important yet poorly controlled variable in language-guided embodied AI. Existing benchmarks typically pair each task with a single static instruction, making it difficult to study how agent behavior changes when the same task is described at different levels of detail. We introduce MINI-BEHAVIOR-GRAN, a new benchmark for controlled studies of instruction granularity that extends MINI-BEHAVIOR with multiple instruction variants per task, ranging from high-level goal descriptions to step-by-step guidance. Using this benchmark, we compare four candidate metrics for cross-task granularity quantification: token count, entity count, action-verb count, and *planning-width*, and find that width correlates most consistently with agent performance. Using width to organize training and evaluation further reveals a non-monotonic U-shaped relationship between instruction granularity and performance, with peaks at both fine and coarse extremes. Further analysis suggests that the coarse-granularity performance rebound is associated with shallow grounding, where agents learn vision-dominant policies.

1 Introduction

Language-guided embodied agents learn a policy $\pi(a|v, l)$ that maps visual observations v and language instructions l to executable actions a . Recent progress has largely focused on model architectures, most notably the Vision-Language-Action (VLA) paradigm (Kim et al., 2024; Black et al., 2025; Shukor et al., 2025) and its advanced variants that adapt *world models* for future frame prediction ($v_t \mapsto v_{t+1}$) (Wu et al., 2026; Zhao et al., 2025; Bi et al., 2025) or *Chain-of-Thought* (CoT) reasoning for intermediate planning (Ye et al., 2025; Lu et al., 2025; Yin et al., 2025). Yet the impact of *instruction granularity* remains underexplored. Instruction granularity refers to the level of detail provided in l to guide agent decision-making (Aru-

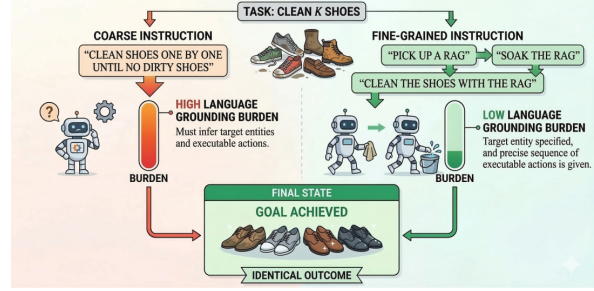


Figure 1: For the same task (cleaning k shoes), a coarse instruction requires the agent to infer target entities and executable actions. In contrast, a fine-grained instruction provides more detailed steps, significantly reducing the language grounding effort to reach the goal.

mugam et al., 2017), and it is typically an uncontrolled variable in current datasets and evaluations. It matters because varying granularity fundamentally alters the cross-modal alignment difficulty and the latent planning burden imposed on the agent. Understanding this impact is crucial not only for optimizing instruction design during training, but also for ensuring model robustness across diverse human instruction styles encountered at test time.

A systematic investigation of instruction granularity has been hindered by two key bottlenecks. First, existing embodied AI benchmarks lack controlled variation in instruction granularity. Typically, each task instance is paired with a single, static instruction, making it difficult to study how an agent’s performance scales as instructions shift from fine-grained (e.g., detailed step-by-step) to coarse (e.g., high-level goal descriptions). For example, popular benchmarks such as LIBERO and BEHAVIOR-1K provide only fixed or high-level instructions. Even recent suites like LIBERO-PLUS (Fei et al., 2025), which aim to increase diversity, primarily vary visual conditions (e.g., lighting, sensor noise) while keeping the language input fixed.

Second, and perhaps more fundamentally, the

field lacks a well-accepted and formalized metric for quantifying *instruction granularity*. Since granularity represents the density of guidance provided for decision-making, an ideal metric should directly reflect the complexity of grounding language into executable actions. With the advent of Large Language Models (LLMs), a seemingly natural proxy is to use the hierarchical depth of an instruction, i.e., its specific level within a decomposed plan, as a measure of granularity (Gao et al., 2025). However, such a metric fails to support meaningful cross-task comparisons because it conflates instructional detail with the task’s inherent complexity. Different tasks exhibit fundamentally different internal structures: some involve interleaved subgoals that resist clean hierarchical decomposition (e.g., the Sussman anomaly; Sussman, 1975), whereas others can be arbitrarily elongated into superficial substeps. Consequently, hierarchical levels cannot establish a consistent granularity scale across diverse tasks, thereby undermining their utility for investigating the general impacts of granularity on language grounding.

To enable systematic study of how instruction granularity shapes language-guided embodied agents, we introduce MINI-BEHAVIOR-GRAN, an extension of MINI-BEHAVIOR (Jin et al., 2023) that provides multiple instruction variants per task, spanning from high-level goal description to step-by-step instructions. Leveraging Minigrad’s engine (Chevalier-Boisvert et al., 2023), we construct 803 unique problem instances across 20 domains, yielding 10,253 trajectory episodes, each paired with dynamic instructions that evolve as the agent progresses through the task. Within this controlled setting, we compare four metrics for quantifying granularity: token count, entity count, action-verb count, and *width*. Unlike *decomposition depth*, *width* measures the max state features jointly tracked per instruction, thereby enabling cross-task comparability of planning complexity (Bonet et al., 2019; Drexler et al., 2024).

Our study shows that *width* correlates more consistently with agent performance across task horizons and VLA model variants than the other three metrics, suggesting that it is a stronger quantitative proxy for instruction granularity in language grounding. Organizing training by *width* reveals a U-shaped relationship between granularity and performance. Probing this effect via instruction predictability ($P(l|v)$) shows that the coarse-granularity rebound is associated with shal-

low grounding, whereby agents tend to bypass multimodal alignment and rely on vision-dominant policies.

Contribution. We make the following contributions: (1) Introduce MINI-BEHAVIOR-GRAN, a benchmark for controlled study of instruction granularity. (2) Demonstrate that *width* is the most consistent cross-task proxy among the metrics we evaluate. (3) Unveil a non-monotonic U-shaped relationship between granularity and agent performance; and (4) Provide evidence that the coarse-granularity performance rebound is linked to shallow grounding.

2 MINI-BEHAVIOR-GRAN: A Controlled Benchmark for Instruction Granularity

Following prior work on varying granularities and abstraction in robot instructions (Arumugam et al., 2019), we define *instruction granularity as the level of details and abstraction with which a language instruction specifies a task. In embodied settings, instruction granularity determines the latent planning complexity that an agent must resolve to map instructions into executable actions.*

Reiterating the example in Figure 1, we note that a coarse instruction (e.g., “clean the shoes”) leaves the agent with a large language grounding gap, as it must infer the target entities (e.g., which shoes to clean) and the executable actions (e.g., how to clean). In contrast, a fine-grained instruction provides detailed steps that significantly reduce the language grounding effort to fulfill the same task. In this section, we describe how we construct MINI-BEHAVIOR-GRAN to systematically vary instruction granularity for the same task.

2.1 Shared Feature Space and Symbolic Layer

Many embodied AI simulators expose symbolic representations alongside raw pixel observations for task specification, progress monitoring, or success evaluation (Shridhar et al., 2020; Puig et al., 2020; Li et al., 2022a,b). We leverage this symbolic backbone to define a shared feature space Φ common across all tasks, representing each environment state s as a feature vector $\phi(s) = (\phi_1(s), \dots, \phi_n(s))$. Features in Φ include both Boolean variables (e.g., H_r : holding the rag; S_r : the rag is soaked) and numerical variables (e.g., p_r : distance to the rag; N_f : number of uncleaned shoes). This shared feature space provides the underlying vocabulary for representing instructions,

which in turn allows us to quantify instruction granularity across tasks.

2.2 Rule-Based Instruction Representation

Building on the shared feature space Φ , we construct each instruction as a set of “if-then” rules that specify desired state progression. This choice is motivated by the fact that sequential decision-making tasks in automated planning are commonly modeled in terms of state variables and transitions with preconditions and effects, following the STRIPS formalism (Haslum et al., 2019). Thus, rule-based instructions explicitly encode the latent planning complexity an instruction imposes, as agents must determine how to transition from states satisfying a rule’s precondition to those satisfying its effect. Concretely, an instruction in this study is a set of rules $\mathcal{R} = \{r_1, \dots, r_m\}$, where each rule r_i is of the form $C_{r_i} \mapsto E_{r_i}$:

- C_{r_i} (the *condition*) is a conjunction of feature constraints. For a Boolean feature p , a constraint is p or $\neg p$; for a numerical feature n , it is $n = 0$ or $n > 0$.
- E_{r_i} (the *effect*) is a conjunction of feature changes. For a Boolean feature p , an effect is p , $\neg p$, or $p?$ (allowing any change); for a numerical feature n , it is $n \downarrow$ (decrease), $n \uparrow$ (increase), or $n?$ (any change).

For each task, we begin by manually constructing a finest-grained rule set $\mathcal{R}^{(0)}$ that decomposes the task into atomic steps, i.e., each rule typically can be fulfilled by a single action. We use the task of *cleaning k shoes* as a running example. Table 1 lists the relevant features from the shared space Φ used in this task.

$\mathcal{R}^{(0)}$ contains over a dozen atomic rules. For brevity, we show representative rules from each phase (full set in § G).

Representative rules from $\mathcal{R}^{(0)}$ (Cleaning Shoes).

$\{\neg H_r, p_r > 0\} \mapsto \{p_r \downarrow\}$	(approach rag)
$\{H_r, \neg T_s, p_s = 0\} \mapsto \{T_s\}$	(turn on tap)
$\{H_r, T_s, p_s = 0\} \mapsto \{S_r\}$	(soak rag)
$\{S_r, N_f > 0, p_f > 0\} \mapsto \{p_f \downarrow\}$	(approach shoe)
$\{S_r, N_f > 0, p_f = 0\} \mapsto \{N_f \downarrow, \neg H_r\}$	(clean and drop)
$\{N_f = 0, \neg O_r, H_r\} \mapsto \{\neg H_r, O_r\}$	(put away rag)

A coarse rule is not created by mechanically concatenating the preconditions and effects of its constituent atomic rules. Instead, we treat each merge as an abstraction of a contiguous subplan

Feature	Description
H_r	holding the rag (Boolean)
p_r	distance to the nearest rag (numerical)
p_s	distance to the sink (numerical)
p_f	distance to the nearest uncleaned shoe (numerical)
N_f	number of uncleaned shoes (numerical)
T_s	tap (sink) is turned on (Boolean)
S_r	rag is soaked (Boolean)
C_i	shoe i is cleaned (Boolean)
O_r	rag is on the floor (Boolean)

Table 1: Features used in the Cleaning Shoes task.

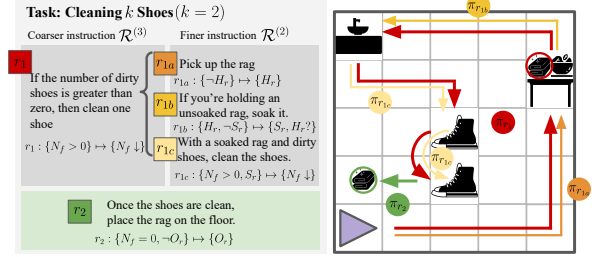


Figure 2: A running example of constructing coarse-grained instructions by merging atomic rules into macro-rules. Features H_r and S_r are omitted as they can be consumed entirely within the subplan chunk.

segment into a macro-rule. The macro-rule exposes only the *net input/output interface* of that segment: its condition specifies when the segment applies, and its effect captures the overall progress achieved. Features that can be consumed entirely within the segment are thus omitted from the expression. As shown in Figure 2, the $\mathcal{R}^{(2)}$ rules for picking up the rag (r_{1a}), soaking it (r_{1b}), and cleaning one shoe (r_{1c}) form a coherent subplan r_1 that abstracts into a single coarse rule in $\mathcal{R}^{(3)}$, which captures the net progress of cleaning one shoe while hiding the intermediate features H_r and S_r .

After each merge, we verify the semantic validity of the resulting coarse rules through expert review and use a planning solver to ensure every $\mathcal{R}^{(k)}$ is well-formed: sequentially reachable and goal-terminating (Bonet and Geffner, 2021).

2.3 Dynamic Instruction Generation

To communicate these rules to the agents, we translate each rule into a natural-language sentence via a template-based generator. The templates verbalize the feature conditions and desired effects in each rule. For example, $\{H_r, \neg S_r\} \mapsto \{S_r\}$ is realized as “If you are holding the rag but the rag is not yet soaked, soak the rag.”

The benchmark supports *dynamic* instruction generation that updates with task progress. As illustrated in Figure 3, during both training and

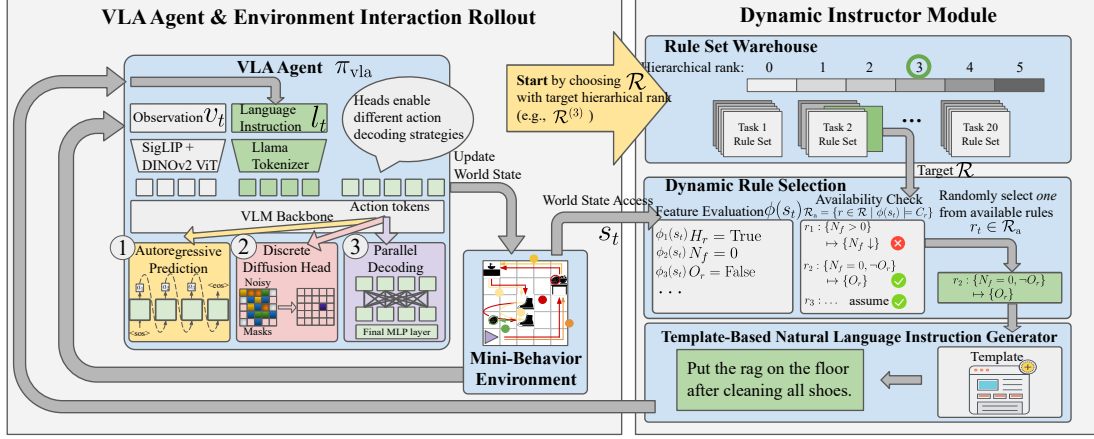


Figure 3: MINI-BEHAVIOR-GRAN Experimental Pipeline. We evaluate the impact of instruction granularity across three VLA action-decoding strategies. During inference time, a dynamic instructor evaluates the current world state to decide available rule set $\mathcal{R}_a$ and active rule r_t . A rule-based natural language instruction l_t is generated accordingly. The process iterates until completion or max steps.

Rank	N	Len	U _{tok}	Plan
0	51,474	40.58 ± 10.59	145	83.43 ± 53.66
1	27,837	38.19 ± 11.53	139	85.81 ± 56.91
2	14,997	33.51 ± 11.17	253	93.68 ± 57.24
3	6,816	34.09 ± 13.99	171	120.26 ± 51.39
4	1,691	41.78 ± 22.67	112	169.32 ± 37.71
5	923	33.31 ± 7.68	85	146.70 ± 23.56

Table 2: Distribution of instruction variants across granularity ranks. N: number of instruction instances; Len: mean instruction length in words; U_{tok}: number of unique tokens; Plan: mean plan length in steps.

evaluation, an instructor module monitors the simulator state at each timestep, computes the set of applicable rules $\mathcal{R}_a$, and determines which rule r_t to activate. This setup ensures that the agent always receives state-relevant guidance, in contrast to existing benchmarks (Li et al., 2022b; Fei et al., 2025) that provide only a single static instruction.

2.4 Benchmark Analysis

Table 2 summarizes the scale and coverage of MINI-BEHAVIOR-GRAN. Across 20 domains, we define a shared feature space Φ with 166 grounded features. Task-specific feature usage ranges from only 2 features in simple tasks such as *Opening Packages* to 29 in long-horizon tasks such as *Cleaning Up the Kitchen*. Despite their structured origin, the resulting instructions are of moderate linguistic complexity, averaging 40.2 words in length (see § F.3 for details).

3 Quantifying Instruction Granularity

While hierarchical rank remains a useful within-task measure, it cannot be used as a consistent

cross-task metric to study how instruction granularity impacts agent performance across a diverse task set. The reason is that the number of attainable levels depends on the internal structure of each task: some tasks admit long chains of cleanly mergeable subgoals, whereas others involve interleaved dependencies that prevent further merging. Consequently, the same hierarchical rank across two different tasks may entail vastly different language grounding complexities. We therefore seek a separate cross-task metric for quantifying granularity.

3.1 Candidate Metrics

To this end, we evaluate four candidate metrics, each corresponding to a different hypothesis about which properties of an instruction plausibly reflect its granularity and associated grounding burden. *Token Count* measures *verbal elaboration*: finer-grained instructions often require more words to spell out procedural details. *Entity Count* measures *referential grounding load*, since mentioning more objects or locations requires the agent to ground a larger set of task-relevant referents. *Action-Verb Count* measures *procedural complexity*, as finer-grained instructions tend to specify more individual actions. While these three metrics capture surface-level properties, our fourth metric, *Width*, captures a more structural notion of planning burden.

3.2 Instantiating Width in MINI-BEHAVIOR-GRAN

Among the candidate metrics we consider, *width* is the least surface-oriented. Originating from width-based planning (Lipovetzky and Geffner, 2012,

State Progression	Concurrent Features Tracked	Width
Initial State	$[\langle \rangle]$	0
Navigate to rag	$[\langle p_r = X \rangle, \dots, \langle p_r = 0 \rangle]$	1
Pick up rag	$[\langle p_r = 0 \rangle, \langle H_r \rangle]$	1
Navigate to sink	$[\langle H_r, p_s = X \rangle, \dots, \langle H_r, p_s = 0 \rangle]$	2
Turn on tap (sink)	$[\langle H_r, p_s = 0 \rangle, \langle T_s \rangle]$	2
Soak rag	$[\langle H_r, T_s, p_s = 0 \rangle, \langle S_r \rangle]$	3
Navigate to Shoe 1	$[\langle H_r, S_r, p_{f1} \downarrow \rangle, \dots]$	3
Clean Shoe 1	$[\langle H_r, S_r, p_{f1} = 0 \rangle, \langle C_1 \rangle]$	3
Navigate to Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} \downarrow \rangle, \dots]$	4
Clean Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} = 0 \rangle, \langle C_1, C_2 \rangle]$	4
Put rag on floor	$[\langle C_1, C_2, H_r \rangle, \dots, \langle C_1, C_2, O_r \rangle]$	3

Table 3: State progression and width for the “Cleaning k Shoes” ($k = 2$). Width scales as $w = k + 2$.

2017), for a rule r_i , its width $w(r_i)$ is the maximum number of features that must be simultaneously tracked along an optimal transition from a state satisfying C_{r_i} to one satisfying E_{r_i} . This differs from simply counting the number of features mentioned in the rule’s condition or effect, since these features need not all be coordinated concurrently. We instantiate instruction-level width by aggregating over its constituent rules $w(\mathcal{R}) = \max_{r_i \in \mathcal{R}} w(r_i)$. A special case of $w = 0$ occurs for situations where the desired state progression can be achieved in zero or one step (Bonet and Geffner, 2024); this will appear in our experimental analysis.

Table 3 illustrates width computation for a two-shoe cleaning task. Approaching the rag ($p_r \downarrow \rightarrow p_r = 0$) requires tracking only p_r ($w = 1$). Obtaining the rag at $p_r = 0$ ($\rightarrow H_r$) also needs only p_r ($w = 1$). Then, maintaining H_r while progressing raises width to 2. Soaking the rag requires simultaneously maintaining three features: at sink ($p_s = 0$), holding rag (H_r), and tap on (T_s), yielding $w = 3$. Cleaning the first shoe (C_1) requires tracking H_r , S_r , and p_f ($w = 3$). Cleaning the second shoe adds maintaining C_1 , increasing width to 4. We acknowledge that w is not a direct measure of internal grounding difficulty, since learning-based embodied agents operate as black-box decision-makers rather than explicit width-based search. However, w provides a cross-task proxy for the latent planning demands. We thus view it as a planning-inspired measure of granularity.

4 Experiments

4.1 Experimental Setup

Data. We use MINI-BEHAVIOR-GRAN, which contains 20 long-horizon household tasks defined over a shared feature space Φ . For each task, the benchmark provides 50 training and 10 evaluation instances, each paired with reference trajectories generated by the BFWS planning solver (Lipovet-

zky and Geffner, 2017).

Action-Decoding Archetypes. The primary architectural difference among recent VLA models lies in their action-decoding strategy (Zhong et al., 2025). We therefore adopt a unified VLM backbone and instantiate three representative action-decoding paradigms (see the left panel of Figure 3): (1) **Autoregressive (AR)**, exemplified by RT-2 (Zitkovich et al., 2023), OpenVLA (Kim et al., 2024), and related models (Reed et al., 2022; Huang et al., 2024; O’Neill et al., 2024), which predict actions sequentially via causal attention; (2) **Discrete Diffusion**, which predicts action chunks through a diffusion process (Black et al., 2025; Bjorck et al., 2025; Shukor et al., 2025; Liang et al., 2025); and (3) **Parallel Decoding**, which uses bidirectional attention for single-pass action prediction (Kim et al., 2025b; Song et al., 2025; Yin et al., 2025; Wang et al., 2025; Xiao et al., 2025). Implementation details are given in § H.

Two-Stage Experimental Protocol. We conduct experiments in two stages. In **Stage I** (§ 4.2), we keep the training distribution fixed and train all models under a mixed-granularity regime that samples from all available instruction variants. This stage is used only to compare candidate granularity metrics by how consistently they correlates with agent Success Rate (SR) across tasks, task horizons, and action-decoding architectures. In **Stage II** (§ 4.3–§ 4.4), after identifying which is the most consistent metric, we use it to partition instruction variants into *Fine*, *Medium*, and *Coarse* groups and study how different granularity mixtures affect learning and generalization. Specifically, we consider seven training settings: **Pure F/M/C** (1:0:0), (2) **Centroid** (1:1:1): uniformly sample from all three classes; (3) **Axial F/M/C** (3:1:1): emphasizes one class with some exposure to the others.

4.2 RQ1: Which metric best captures instruction granularity?

We first ask which candidate metric most consistently captures instruction granularity in a way that it reflects embodied language grounding difficulty across tasks. Following the Stage-I protocol in § 4.1, we train all models under a balanced granularity regime and evaluate the Pearson correlation between each candidate metric and agent SR. A useful cross-task metric should exhibit a stable relationship with performance across tasks of different sequential burdens, rather than only within a

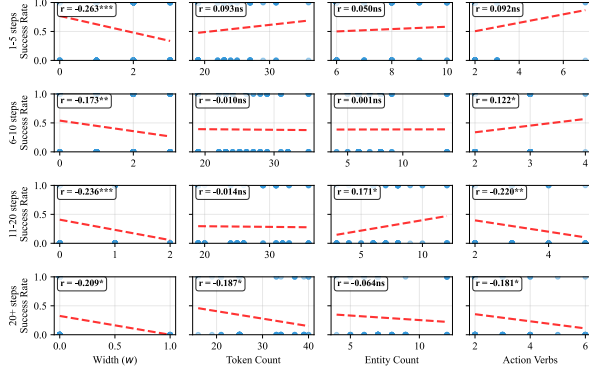


Figure 4: Metric-success correlation by task-horizon, with *width* showing consistent negative correlations.

narrow subset of the benchmark.

Why partition by instruction horizon? A critical confound in evaluating instruction granularity is sequential burden. A natural choice would be to stratify examples by the total number of primitive actions in the expert trajectory. However, this is not the most relevant notion of difficulty in our setting. In MINI-BEHAVIOR-GRAN, agents do not ground a single static instruction over the entire trajectory; instead, the instructor issues a sequence of rule-conditioned instruction updates as execution unfolds. The more relevant notion of sequential burden is therefore the *instruction horizon*, defined as the number of rule calls issued during execution. We thus partition metric–SR associations evaluation instances by instruction horizon.

Width vs. surface-level metrics. We compare between token count, entity count, action-verb count, and width. Figure 4 show that width is the only metric that exhibits a statistically significant and consistently negative correlation with SR across all instruction-horizon bins (e.g., -0.263 in the 1–5 steps bin). In other words, as width increases, agent success decreases in a stable manner regardless of sequential burden. By contrast, the other three metrics are notably less consistent. Token count is insignificant in the first three bins and becomes significantly negative only in the longest-horizon bin. Entity count is mostly insignificant and even becomes weakly positive in the 11–20 bin. Action-verb count reaches significance in several bins, but its direction flips across horizons, indicating that it does not provide a stable notion of granularity across tasks. We thus use width to organize training in the remainder of the paper.

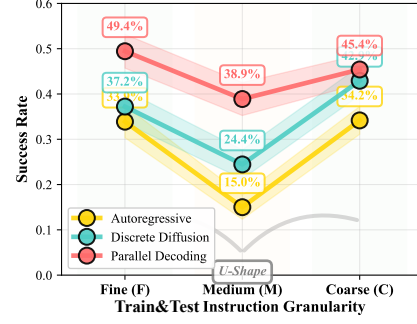


Figure 5: In-Distribution (matched train/test) SR across granularities (i.e., train Fine $\rightarrow$ test Fine, etc). A non-monotonic U-shaped trend appears.

4.3 RQ2: How does agent performance vary across instruction granularity?

Since RQ1 identifies width as the most consistent cross-task proxy for instruction granularity, we use it to organize the remaining experiments. To enable tractable analysis while maintaining statistical power, we partition instruction variants into three granularity groups based on width: *Fine* ($w = 0$), *Medium* ($w = 1$), and *Coarse* ($w \geq 2$). This grouping is necessary because higher widths ($w \geq 3$) have fewer instances and appear in fewer domains, making per-width analysis underpowered for cross-training comparisons. However, to ensure this grouping does not obscure finer-grained trends, we present a per-width breakdown of performance in Table 5. Figure 5 reveals a consistent non-monotonic U-shaped relationship between instruction granularity and agent performance across all three action-decoding archetypes. At the fine extreme ($w = 0$), where instructions require minimal latent reasoning, models achieve high success rates. Performance then drops at medium granularity ($w = 1$), before rebounding under coarse instructions ($w \geq 2$). This is notable because width tracks latent coordination burden: if granularity affected performance only through increasing planning difficulty, one would expect a monotonic decline rather than a recovery at the coarse end.

While our main analysis groups $w \geq 2$ as *Coarse*, examining per-width performance reveals a graded structure within this regime. Table 5 shows that SR at $w = 2$ and $w = 3$ declines progressively, collapsing entirely at $w \geq 4$. Nevertheless, the overall *Coarse* group still outperforms *Medium* ($w = 1$) when aggregated across tasks. This aggregation is dominated by $w = 2$ (which appears in 19 tasks), while the lower-performing $w = 3$ and $w \geq 4$ appear in only 10 and 3 tasks

respectively, so their inclusion does not reverse the coarse-medium ordering.

4.4 RQ3: What drives the U-shaped effect?

The coarse-granularity rebound is initially counter-intuitive: if width reflects latent coordination burden, why does performance improve again when instructions become coarser? We hypothesize that this rebound stems from increasing instruction predictability $P(l|v)$, which incentivizes agents to bypass language in favor of visuo-motor shortcuts.

This increased predictability has a formal basis: coarse instructions subsume multiple finer-grained variants. Formally, consider a surjective mapping f from fine-grained labels l_i to coarser ones c_j . The conditional probability of a coarse label aggregates its fine-grained constituents: $P(c_j|v) = \sum_{l_i \in f^{-1}(c_j)} P(l_i|v)$. This aggregation makes coarse instructions statistically easier to predict from vision alone.

We confirm this empirically by training a VLM-based instructor (Qwen3-VL-4B with LoRA) to predict instructions from visual observations. As Figure 6 shows, cross-entropy loss decreases monotonically with coarseness (Fine: 0.078, Medium: 0.063, Coarse: 0.047), a relative drop of 40% from fine to coarse, confirming that $P(l|v)$ indeed increases with coarseness.

We next ask whether this increased predictability changes how much models actually rely on language. Following prior work (Fei et al., 2025), we use Δ SR when language is weakened or removed as a probe of grounding reliance. When full instructions are removed entirely, fine-trained policies suffer a severe 36.3% drop, whereas coarse-trained policies show a much smaller 20.0% gap (Figure 7). Notably, under reduced informativeness (*w/ goal lang*), the Pure-C policy exhibits only a negligible 2.8% advantage over having no language at all (15.6% vs. 12.8%). This indicates that policies trained on coarse instructions develop shallow grounding: language is not deeply integrated into decision-making.

Table 4 reveals that action token attention supports the shallow grounding hypothesis. Using a layer-wise Binomial test ($H_0 = 0.5$), we find Discrete Diffusion models significantly shift toward visual dominance at the coarse end (T2, $p = 0.01$; T3, $p < 0.01$). AR models mirror this staged behavior, exhibiting a highly significant language grounding phase (T1: 25/32, $p \approx 0.001$) and a directional shift toward visual reliance (T3: 21/32).

Conversely, Parallel models show no significant patterns; we attribute this to the early modality entanglement inherent in bidirectional attention (?), which prevents the emergence of staged, layer-wise modality transitions.

Together, these results reveal a three-regime account of the U-shaped performance pattern. At **fine granularity** ($w = 0$), Low $P(l|v)$ (high CE loss) means language provides unique information. Combined with low-width grounding burden, this drives genuine multimodal alignment for generating actions, yielding high SR. **Medium granularity** ($w = 1$) introduces a moderate increase in grounding burden. The model attempts to ground language but struggles with the increased complexity, leading to the lowest performance. At **coarse granularity** ($w \geq 2$), instructions are too complex to ground effectively, and high $P(l|v)$ incentivizes agents to exploit visual cues. This visual-dominant strategy recovers SR but sacrifices language grounding.

Grounding via Mixed-Granularity Training.

We hypothesize that increasing the diversity of the training language space should artificially lower average $P(l|v)$, forcing agents to re-engage with text. Our mixed-granularity training confirms this: all mixed-granularity models exhibit substantially larger Δ SR gaps than their Pure-X counterparts. Among these, the fine-weighted mixture *Axial-F* emerges as the optimal compromise, achieving strong overall performance while maintaining larger Δ SR gaps indicative of language grounding.

Asymmetric Generalization. We observe an asymmetric transfer pattern (Table 5): coarse-trained models successfully zero-shot transfer to fine-grained instructions, but fine-trained models fail on coarse-grained ones. This asymmetry aligns with our Δ SR analysis: coarse training induces a visual-dominant policy, hence refining the instruction yields minimal behavioral change. Fine training induces tight coupling with language, making the policy brittle when detailed guidance drops. Also see § I for failure mode analysis.

5 Related Work

Instruction Abstraction and Granularity. Here, *instruction granularity* refers to the level of detail that affects the latent planning burden of an instruction. This differs from *linguistic abstraction*, defined as representational compression (Wing, 2010;

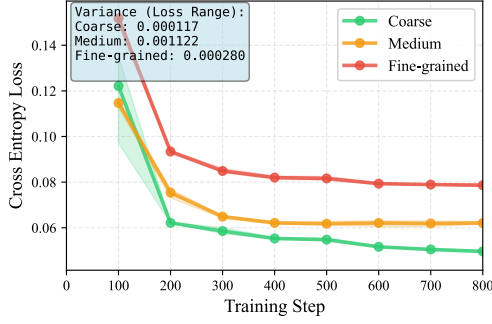


Figure 6: VLM instruction-prediction Cross Entropy (CE) loss across granularities and 3 seeds. Larger CE loss indicates lower $P(l | v)$.

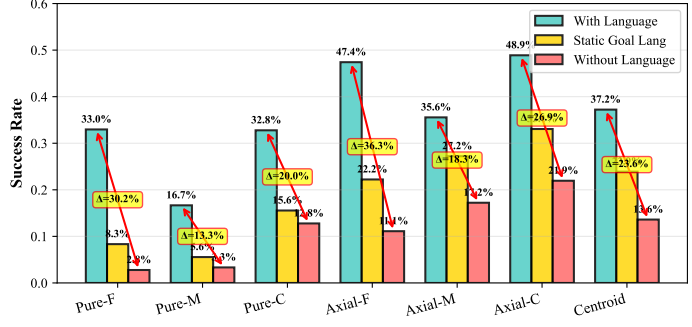


Figure 7: Smaller SR gaps indicate a shift towards shallow grounding. Under semantic perturbation, the Pure-C setting exhibits tiny variance (2.8%) between w/ *goal lang* and w/o *lang*.

Model	T1	T2	T3
AR	78% (25/32)**	13% (4/32)	66% (21/32)
Parallel	16% (5/32)	28% (9/32)	16% (5/32)
Diffusion	53% (17/32)	72% (23/32)*	75% (24/32)**

Table 4: Layer-wise (32) attention trends derived from action tokens. Three trends: **T1** (lang attn $\uparrow$ from $F \rightarrow M$), **T2** (lang attn $\downarrow$ from $M \rightarrow C$), and **T3** (vis attn $\uparrow$ from $M \rightarrow C$). Stats significance is denoted by * $p < 0.05$ and ** $p < 0.01$ (See § I.2 for full table). Diffusion and AR models’ T1 and T2/T3 trends supports the hypothesis that *C lang* incentivizes shallow grounding.

Training Setup	Width 0 (20 tasks)	Width 1 (20 tasks)	Width 2 (19 tasks)	Width 3 (10 tasks)	Width ≥ 4 (3 tasks)
Pure-F	34.9%	16.9%	10.8%	1.0%	0.0%
Pure-M	12.2%	22.0%	11.7%	0.0%	0.0%
Pure-C	26.3%	33.3%	32.9%	22.9%	0.0%
Axial-F	50.2%	53.3%	48.8%	34.3%	0.0%
Axial-M	44.7%	43.5%	45.8%	29.5%	0.0%
Axial-C	55.7%	52.2%	50.8%	37.1%	0.0%
Centroid (Mix)	42.7%	38.4%	40.8%	28.6%	0.0%

Table 5: Success rate (%) across training setups and test granularities. Numbers in parentheses indicate task coverage at each width. For Pure models, cells where test width was unseen during training measure zero-shot transfer. Transfer is asymmetric: coarse $\rightarrow$ fine succeeds.

Lachmy et al., 2022; Zheng et al., 2024). Granularity varies independently: “tidy the room” is coarse yet non-abstract (underspecified actions, no compression); “repeat wiping until clean” is specific yet abstract (prescribes action via looped condition).

While not framed as the term “granularity”, we acknowledge that a rich body of work has focused on augmenting language specificity and its impact, for example through coarse-to-fine semantic parsing (Dong and Lapata, 2018; Li et al., 2020). Other work generates compositional instructions via constraint dropout and sub-task recombination (Huang et al., 2025b), or construct counterfactual instructions to augment diversity (Glossop et al., 2025).

Varying instruction detail is often handled via a simple two-part decomposition: high-level plans are converted into sequences of low-level steps before being grounded into actions (Arumugam et al., 2017; Yang et al., 2024; Shi et al., 2025). Most recent Vision Language Action (VLA) works, however, operate with static language instructions (Ma et al., 2025; Li et al., 2025). Some models like InstructVLA and ThinkAct explore having chain-of-thought reasoning or planning steps before action decoding (Huang et al., 2022; Yang et al., 2025; Huang et al., 2025a). These hierarchical defini-

tions remain task-specific and qualitative, lacking a unified, cross-task metric to establish shared granularity levels across tasks. Consequently, quantitatively controlled studies on the formal dynamics of instruction granularity remain scarce.

Language Grounding vs. Shortcuts. Prior work highlights a dichotomy between fragile language sensitivity (Akula et al., 2022) and visuo-motor shortcuts (Xing et al., 2025; Zhang et al., 2026). We reconcile this via width (w), providing complementary insights into the trade-offs between language grounding and visual reliance.

6 Conclusion

We introduce MINI-BEHAVIOR-GRAN, the first testbed for studying the impact of instruction granularity. Comparing four metrics, we find *width* most consistently correlates with grounding difficulty. Using width to organize training reveals a U-shaped granularity-performance pattern, driven by a trade-off between instruction predictability ($P(l|v)$) and planning complexity. We demonstrate that mixed-granularity training balances language sensitivity without sacrificing the visual reasoning skills learned from coarse instructions.

7 Limitations

Discrete Action Spaces. We evaluate our framework in environments with discrete action spaces, a common abstraction in language-conditioned embodied AI (e.g., ALFWorld-based agent (Xiong et al., 2025; Kim et al., 2025a)) and LLM-based agent systems (Qin et al., 2024; Patil et al., 2024), where high-level reasoning is decoupled from low-level execution via skill primitives or API calls. This choice is intentional: it isolates symbolic planning from continuous control noise, allowing us to cleanly attribute observed effects to instruction granularity. However, extending this granularity-aware framework to hybrid discrete-continuous or continuous action spaces remains essential for real-world robotics applications, where low-level motor commands interact with high-level linguistic abstractions.

Scope of Granularity Definition. We view subgoal decomposition as the primary axis of instruction granularity, as it directly determines the latent planning burden—the central construct we aim to measure. This focus aligns with the classical planning view that hierarchical task decomposition is the fundamental mechanism for reducing search complexity. Other forms of granularity (e.g., vagueness, abstraction via loops or branching) are important but introduce additional complexities that would confound our initial investigation. By first isolating and understanding the effect of subgoal decomposition in a controlled setting, we establish a foundation for future work to incorporate these other dimensions. Our benchmark and findings provide a baseline against which such extensions can be compared.

Practicality of width computation. Computing width currently requires access to symbolic state features and optimal plans, which limits its direct applicability to natural language instructions in unstructured environments. We emphasize, however, that width is designed as an analytic metric for controlled studies, not as a deployable measure for real-world settings. Within our benchmark, width enables precise quantification of latent planning burden, allowing us to isolate the effect of instruction granularity and uncover fundamental phenomena such as the U-shaped performance curve and asymmetric transfer. Extending width estimation to realistic settings remains future work and does not affect the validity of our findings.

References

- Arjun R. Akula, Spandana Gella, Aishwarya Padmakumar, Mahdi Namazifar, Mohit Bansal, Jesse Thomason, and Dilek Hakkani-Tur. 2022. ALFRED-L: investigating the role of language for action learning in interactive visual environments. In *EMNLP*, pages 9369–9378. Association for Computational Linguistics.
- Dilip Arumugam, Siddharth Karamcheti, Nakul Gopalan, Edward C. Williams, Mina Rhee, Lawson L. S. Wong, and Stefanie Tellex. 2019. Grounding natural language instructions to semantic goal representations for abstraction and generalization. *Auton. Robots*, 43(2):449–468.
- Dilip Arumugam, Siddharth Karamcheti, Nakul Gopalan, Lawson LS Wong, and Stefanie Tellex. 2017. Accurately and efficiently interpreting human-robot instructions of varying granularities. *arXiv preprint arXiv:1704.06616*.
- Hongzhe Bi, Hengkai Tan, Shenghao Xie, Zeyuan Wang, Shuhe Huang, Haitian Liu, Ruowen Zhao, Yao Feng, Chendong Xiang, Yinze Rong, Hongyan Zhao, Hanyu Liu, Zhizhong Su, Lei Ma, Hang Su, and Jun Zhu. 2025. *Motus: A unified latent action world model*. *Preprint*, arXiv:2512.13030.
- Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, and 1 others. 2025. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*.
- Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, brian ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, and 16 others. 2025. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 17–40. PMLR.
- Blai Bonet, Guillem Francès, and Hector Geffner. 2019. Learning features and abstract actions for computing generalized plans. In *AAAI*, pages 2703–2710. AAAI Press.
- Blai Bonet and Hector Geffner. 2021. General policies, representations, and planning width. In *AAAI*, pages 11764–11773. AAAI Press.
- Blai Bonet and Hector Geffner. 2024. General policies, subgoal structure, and planning width. *Journal of Artificial Intelligence Research*, 80:475–516.
- Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo De Lazcano Perez-Vicente, Lucas Willems, Salem Lahlou, Suman Pal, Pablo Samuel Castro, and J K Terry. 2023. *Minigrid & miniworld: Modular &*

717	customizable reinforcement learning environments	Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky	771
718	for goal-oriented tasks. In <i>Thirty-seventh Conference</i>	Liang, Pete Florence, Andy Zeng, Jonathan Tompson,	772
719	on Neural Information Processing Systems Datasets	Igor Mordatch, Yevgen Chebotar, Pierre Sermanet,	773
720	and Benchmarks Track.	Tomas Jackson, Noah Brown, Linda Luu, Sergey	774
		Levine, Karol Hausman, and Brian Ichter. 2022. In-	775
721	Li Dong and Mirella Lapata. 2018. Coarse-to-fine de-	inner monologue: Embodied reasoning through plan-	776
722	coding for neural semantic parsing. In <i>Proceedings</i>	ning with language models. In <i>Conference on Robot</i>	777
723	of the 56th Annual Meeting of the Association for	Learning, CoRL 2022, 14-18 December 2022, Auck-	778
724	Computational Linguistics (Volume 1: Long Papers),	land, New Zealand, volume 205 of <i>Proceedings</i>	779
725	pages 731–742, Melbourne, Australia. Association	of Machine Learning Research, pages 1769–1782.	780
726	for Computational Linguistics.	PMLR.	781
727	Dominik Drexler, Jendrik Seipp, and Hector Geffner.	Emily Jin, Jiaheng Hu, Zhuoyi Huang, Ruohan Zhang,	782
728	2024. Expressing and exploiting subgoal structure	Jiajun Wu, Li Fei-Fei, and Roberto Martín-Martín.	783
729	in classical planning using sketches. <i>J. Artif. Intell.</i>	2023. Mini-BEHAVIOR: A procedurally generated	784
730	<i>Res.</i> , 80.	benchmark for long-horizon decision-making in em-	785
		bodied AI. In <i>NeurIPS 2023 Workshop on General-</i>	786
731	Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun	ization in Planning.	787
732	Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang,		
733	Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng	Siddharth Karamcheti, Suraj Nair, Ashwin Balakrishna,	788
734	Qiu. 2025. Libero-plus: In-depth robustness anal-	Percy Liang, Thomas Kollar, and Dorsa Sadigh. 2024.	789
735	ysis of vision-language-action models. <i>Preprint</i> ,	Prismatic vlms: Investigating the design space of	790
736	arXiv:2510.13626.	visually-conditioned language models. In <i>Forty-</i>	791
		first International Conference on Machine Learning,	792
737	Chongkai Gao, Zixuan Liu, Zhenghao Chi, Junshan	ICML 2024, Vienna, Austria, July 21-27, 2024.	793
738	Huang, Xin Fei, Yiwen Hou, Yuxuan Zhang, Yudi		
739	Lin, Zhirui Fang, and Lin Shao. 2025. VLA-OS:	Jeonghye Kim, Sojeong Rhee, Minbeom Kim, Do-	794
740	Structuring and dissecting planning representations	hyung Kim, Sangmook Lee, Youngchul Sung, and	795
741	and paradigms in vision-language-action models. In	Kyomin Jung. 2025a. ReflAct: World-grounded deci-	796
742	<i>The Thirty-ninth Annual Conference on Neural Infor-</i>	sion making in LLM agents via goal-state reflection.	797
743	<i>mation Processing Systems</i> .	In <i>Proceedings of the 2025 Conference on Empiri-</i>	798
		<i>cal Methods in Natural Language Processing</i> , pages	799
744	Catherine Glossop, William Chen, Arjun Bhorkar,	33433–33465, Suzhou, China. Association for Com-	800
745	Dhruv Shah, and Sergey Levine. 2025. Cast:	putational Linguistics.	801
746	Counterfactual labels improve instruction follow-		
747	ing in vision-language-action models. <i>Preprint</i> ,	Moo Jin Kim, Chelsea Finn, and Percy Liang.	802
748	arXiv:2508.13446.	2025b. Fine-tuning vision-language-action mod-	803
		els: Optimizing speed and success. <i>arXiv preprint</i>	804
749	Danijar Hafner. 2021. Benchmarking the spectrum of	<i>arXiv:2502.19645</i> .	805
750	agent capabilities. <i>arXiv preprint arXiv:2109.06780</i> .		
		Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti,	806
751	Patrik Haslum, Nir Lipovetzky, Daniele Magazzeni,	Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael	807
752	Christian Muise, Ronald Brachman, Francesca Rossi,	Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan	808
753	and Peter Stone. 2019. <i>An introduction to the</i>	Vuong, Thomas Kollar, Benjamin Burchfiel, Russ	809
754	<i>planning domain definition language</i> , volume 13.	Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang,	810
755	Springer.	and Chelsea Finn. 2024. Openvla: An open-source	811
		vision-language-action model. In <i>CoRL</i> , volume 270	812
756	Chi-Pin Huang, Yueh-Hua Wu, Min-Hung Chen,	of <i>Proceedings of Machine Learning Research</i> , pages	813
757	Yu-Chiang Frank Wang, and Fu-En Yang. 2025a.	2679–2713. PMLR.	814
758	Thinkact: Vision-language-action reasoning via		
759	reinforced visual latent planning. <i>Preprint</i> ,	Royi Lachmy, Valentina Pyatkin, Avshalom Manevich,	815
760	arXiv:2507.16815.	and Reut Tsarfaty. 2022. Draw me a flower: Process-	816
		ing and grounding abstraction in natural language.	817
761	Hui Huang, Jiaheng Liu, Yancheng He, Shilong Li, Bing	<i>Trans. Assoc. Comput. Linguistics</i> , 10:1341–1356.	818
762	Xu, Conghui Zhu, Muyun Yang, and Tiejun Zhao.		
763	2025b. Musc: Improving complex instruction fol-	Chengshu Li, Fei Xia, Roberto Martín-Martín, Michael	819
764	lowing with multi-granularity self-contrastive train-	Lingelbach, Sanjana Srivastava, Bokui Shen, Kent El-	820
765	ing. In <i>ACL (1)</i> , pages 10667–10686. Association for	liott Vainio, Cem Gokmen, Gokul Dharan, Tanish	821
766	Computational Linguistics.	Jain, Andrey Kurenkov, Karen Liu, Hyowon Gweon,	822
		Jiajun Wu, Li Fei-Fei, and Silvio Savarese. 2022a.	823
767	Jiangyong Huang, Silong Yong, Xiaojian Ma, Xiongkun	igibson 2.0: Object-centric simulation for robot learn-	824
768	Linghu, Puhao Li, Yan Wang, Qing Li, Song-Chun	ing of everyday household tasks. In <i>Proceedings of</i>	825
769	Zhu, Baoxiong Jia, and Siyuan Huang. 2024. An	<i>the 5th Conference on Robot Learning</i> , volume 164	826
770	embodied generalist agent in 3d world. In <i>ICML</i> .	of <i>Proceedings of Machine Learning Research</i> , pages	827
		455–465. PMLR.	828

829	Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, Michael Lingelbach, Jiankai Sun, Mona Anvari, Minjune Hwang, Manasi Sharma, Arman Aydin, Dhruva Bansal, Samuel Hunter, Kyu-Young Kim, Alan Lou, Caleb R. Matthews, and 9 others. 2022b. BEHAVIOR-1K: A benchmark for embodied AI with 1,000 everyday activities and realistic simulation. In <i>CoRL</i> , volume 205 of <i>Proceedings of Machine Learning Research</i> , pages 80–93. PMLR.	886
830		887
831		888
832		889
833		890
834		891
835		
836		892
837		893
838		894
839		
840	Wei Li, Renshan Zhang, Rui Shao, Jie He, and Liqiang Nie. 2025. Cogvla: Cognition-aligned vision-language-action model via instruction-driven routing & sparsification. <i>Advances in neural information processing systems</i> .	
841		
842		
843		
844		
845		895
846		896
847		897
848		898
849		899
850		
851		900
852		901
853		902
854		903
855		904
		905
		906
856		
857		907
858		908
859		909
		910
		911
		912
		913
		914
860		915
861		916
862		917
863		918
		919
		920
		921
864		922
865		923
866		924
867		925
868		926
869		927
870		928
871		
872		929
873		930
874		931
875		932
876		933
877		934
878		
879		935
880		936
881		937
882		938
		939
		940
883		
884		941
885		942

